

Gym Management System

Data Dirtying Documentation — Phase 2

1. OVERVIEW & METHODOLOGY

To simulate a realistic raw dataset for the Multi-Stage Cleansing phase, a controlled, reproducible script (`make_dirty_data.py`) was used to inject known data-quality issues into the clean `workout_session` table only. All other tables (Member, Subscription Plan, Gym Branch, Facility) were left unmodified. The process uses a fixed random seed (seed = 42), making every injected error fully reproducible and auditable.

Source table	workout_session
Original row count	300,000
Final row count (after duplicates)	304,500
Random seed	42
Output file	workout_session_dirty.csv

2. INJECTED DATA QUALITY ISSUES

Each issue below was applied to an independently sampled subset of rows, so percentages may overlap across issues (a single row can be affected by more than one issue).

#	Issue	Column Affected	Rows	%	Cleaning Stage
1	Missing values (NULL)	calories_burned	9,000	3.0%	Python
2	Missing values (NULL)	checkout_time	6,000	2.0%	Excel
3	Duplicate rows	all columns	4,500	1.5%	SQL
4	Inconsistent casing / whitespace	workout_type	15,225	5.0%	SQL
5	Spelling typos (e.g. Yogaa, Crosfit)	workout_type	6,090	2.0%	Excel

#	Issue	Column Affected	Rows	%	Cleaning Stage
6	Inconsistent date format (MM/DD/YYYY)	checkin_time	9,135	3.0%	Excel
7	Outlier / invalid values (negative, 9999+)	calories_burned	3,045	1.0%	Python
8	Logical error: checkout before checkin	checkout_time	2,980	1.0%	SQL
9	Extra leading/trailing whitespace	user_id	6,090	2.0%	SQL

Legend — **SQL**: resolved in Stage 1 (Partial SQL Clean) | **Excel**: resolved in Stage 2 (Excel & Power Query) | **Python**: resolved in Stage 3 (Python — 100% Clean)